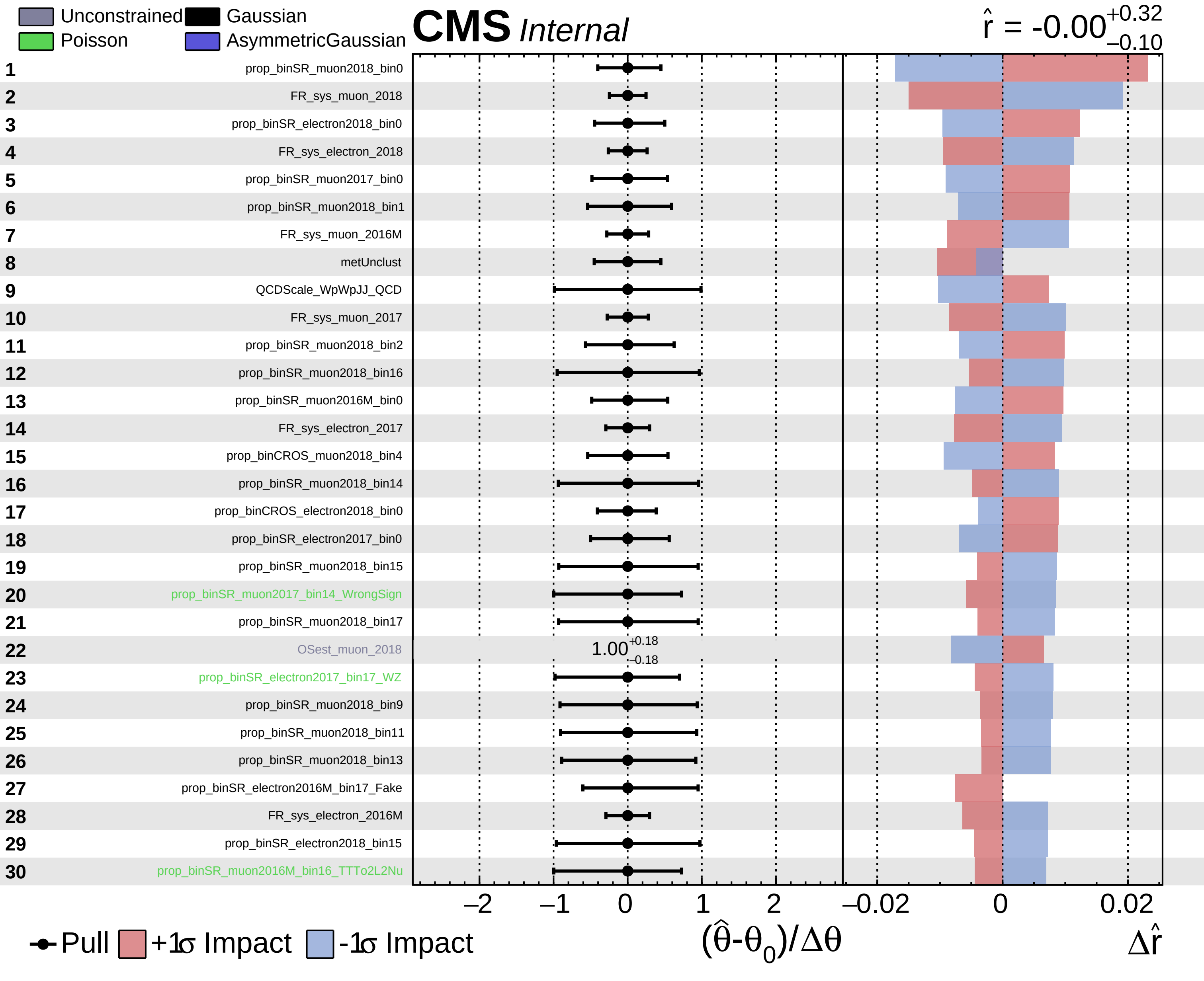
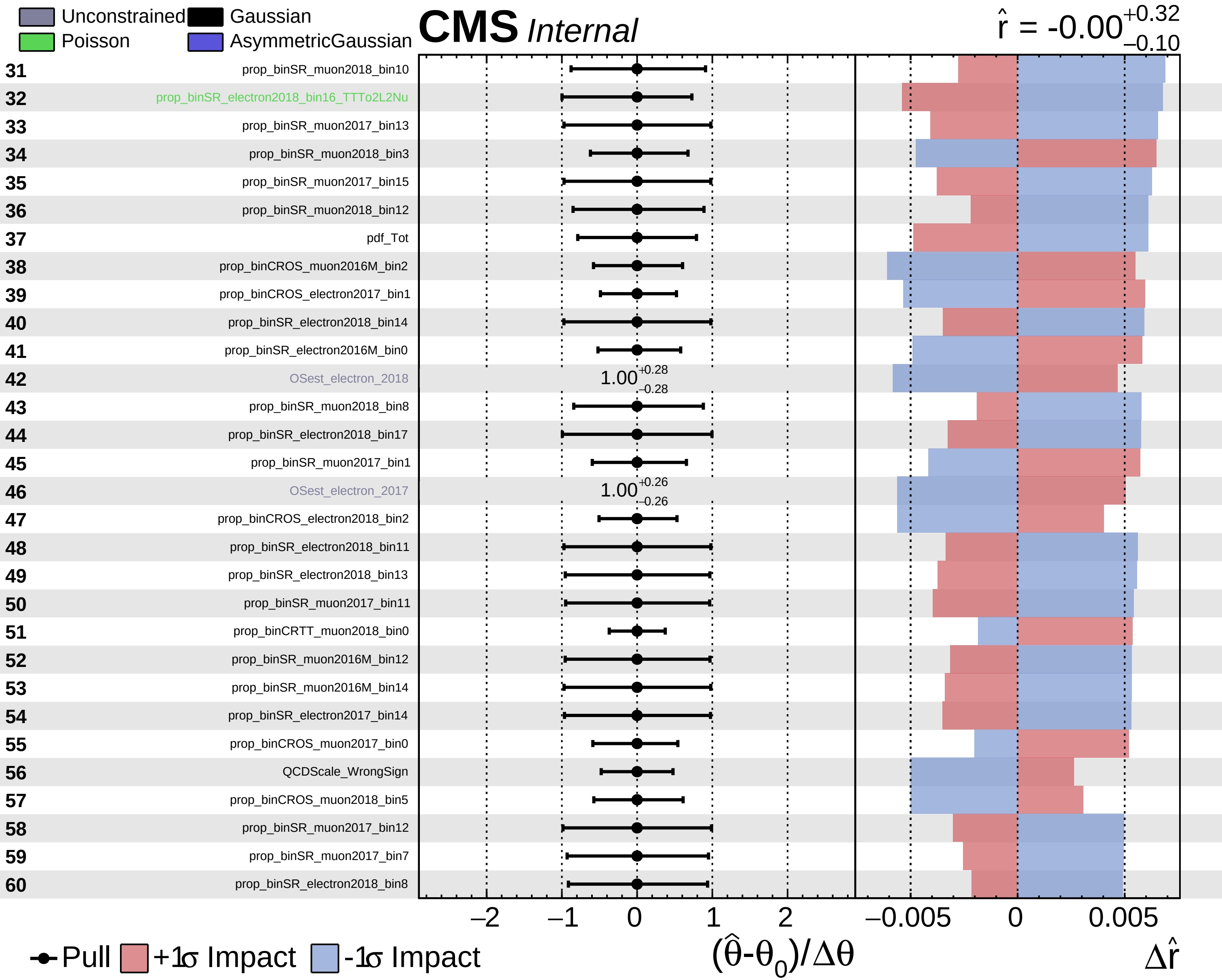
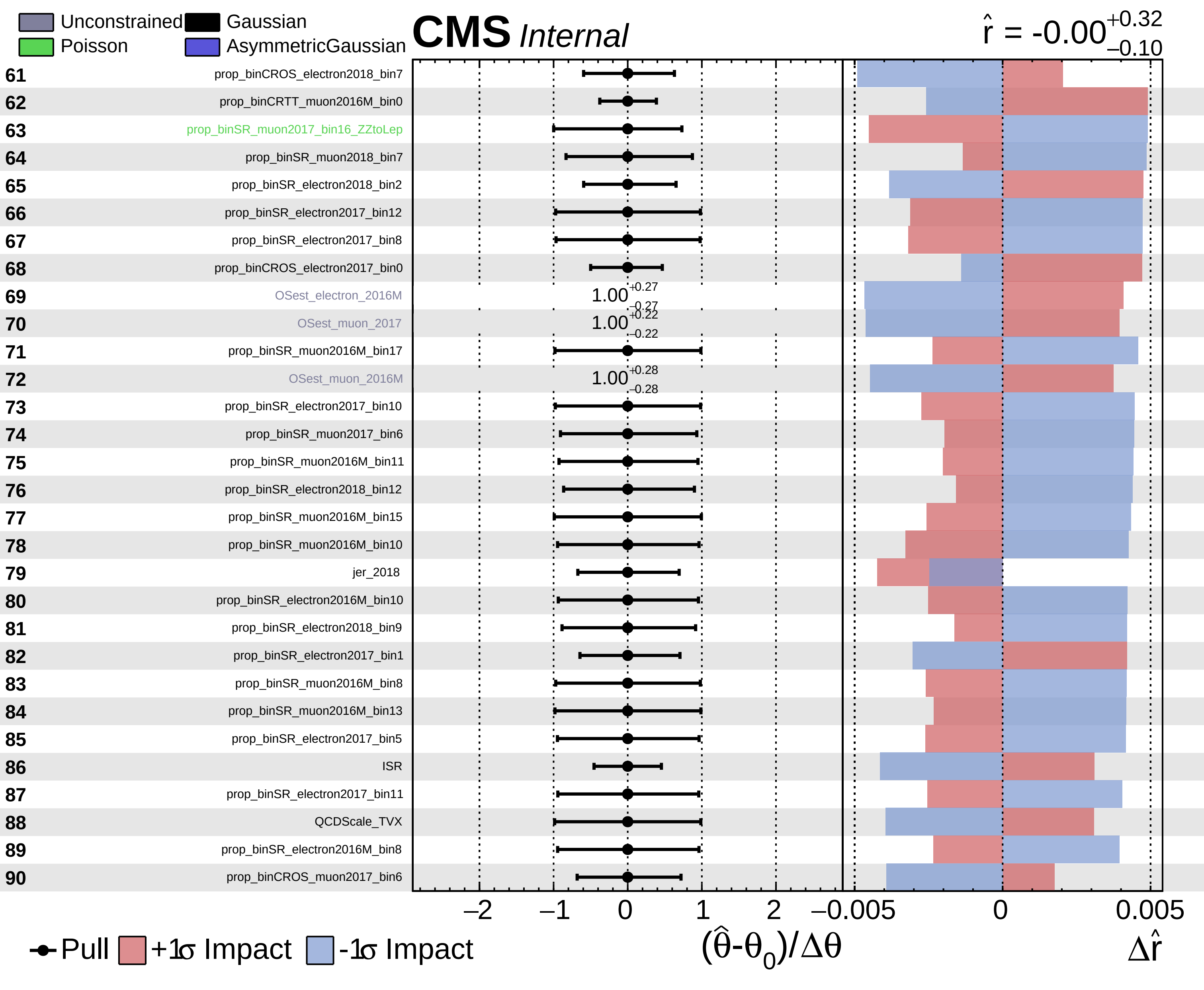
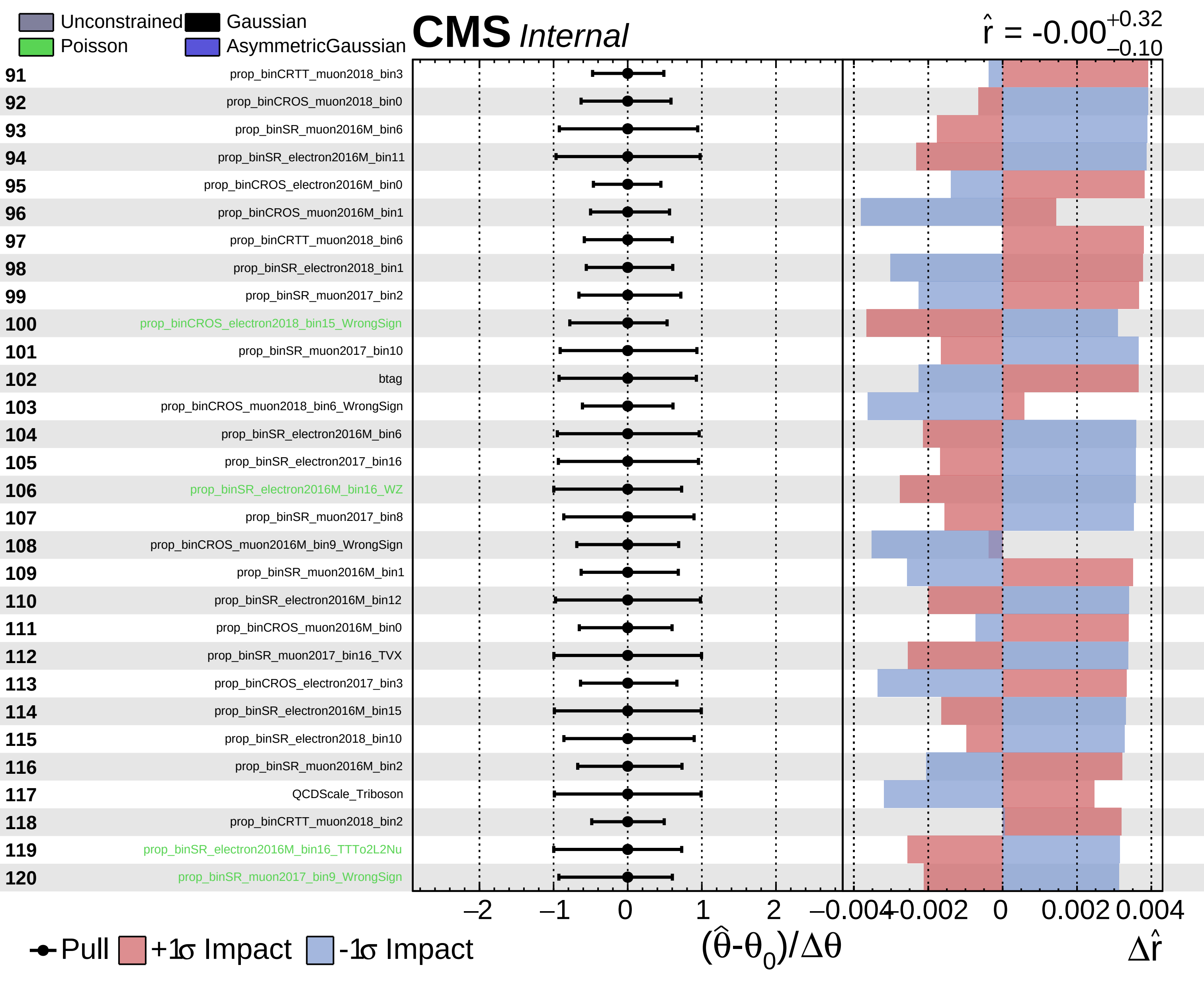


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.32}_{-0.10}$
1 prop_binSR_muon2018_bin0
2 FR_sys_muon_2018
3 prop_binSR_electron2018_bin0
4 FR_sys_electron_2018
5 prop_binSR_muon2017_bin0
6 prop_binSR_muon2018_bin1
7 FR_sys_muon_2016M
8 metUnclust
9 QCDScale_WpWpJJ_QCD
10 FR_sys_muon_2017
11 prop_binSR_muon2018_bin2
12 prop_binSR_muon2018_bin16
13 prop_binSR_muon2016M_bin0
14 FR_sys_electron_2017
15 prop_binCROS_muon2018_bin4
16 prop_binSR_muon2018_bin14
17 prop_binCROS_electron2018_bin0
18 prop_binSR_electron2017_bin0
19 prop_binSR_muon2018_bin15
20 prop_binSR_muon2017_bin14_WrongSign
21 prop_binSR_muon2018_bin17
22 OSest_muon_2018
$1.00^{+0.18}_{-0.18}$
23 prop_binSR_electron2017_bin17_WZ
24 prop_binSR_muon2018_bin9
25 prop_binSR_muon2018_bin11
26 prop_binSR_muon2018_bin13
27 prop_binSR_electron2016M_bin17_Fake
28 FR_sys_electron_2016M
29 prop_binSR_electron2018_bin15
30 prop_binSR_muon2016M_bin16_TTTo2L2Nu
−2
−1
0
1
2
−0.02
0
0.02
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

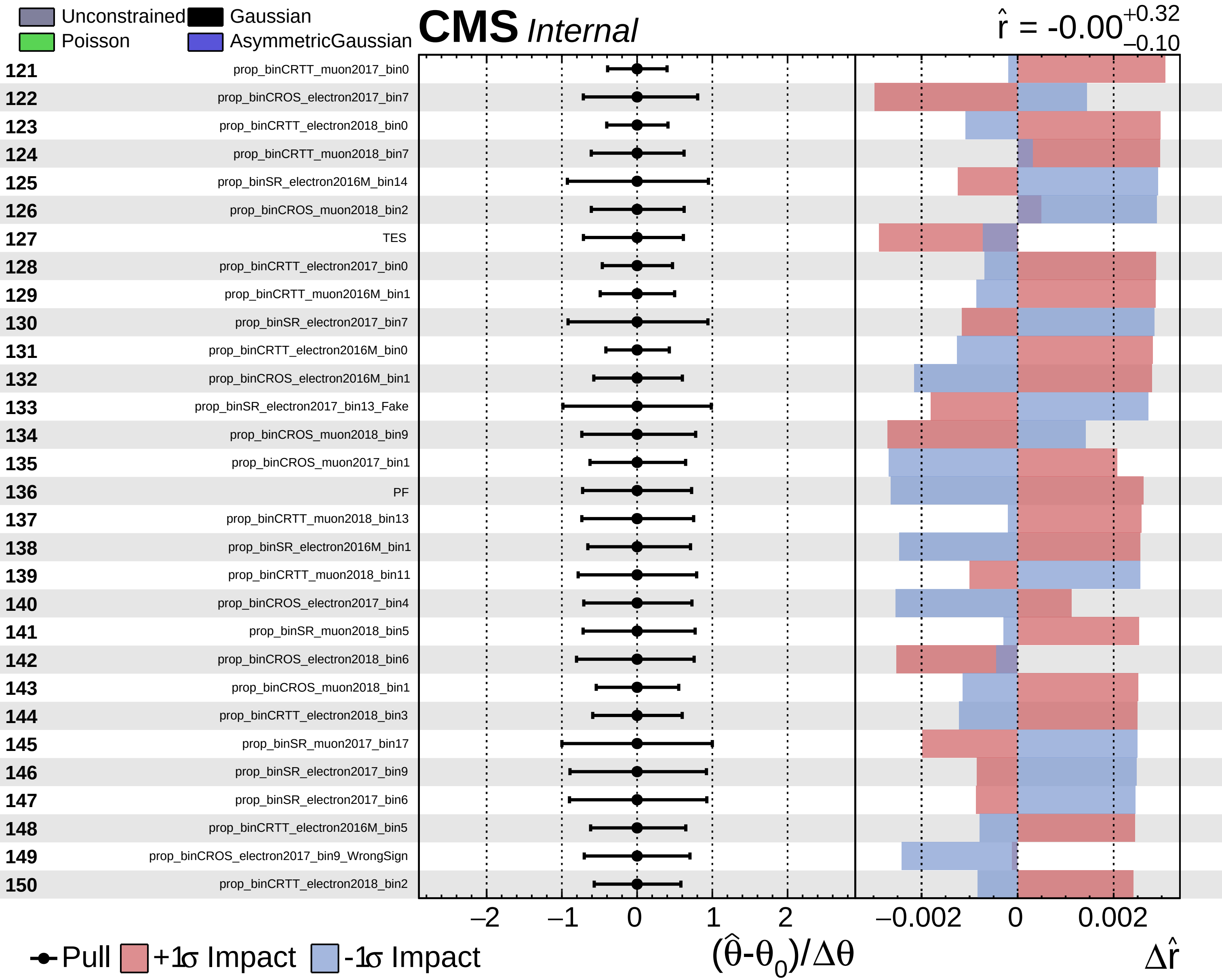


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = -0.00 +0.32 −0.10
31 prop_binSR_muon2018_bin10
32 prop_binSR_electron2018_bin16_TTTo2L2Nu
33 prop_binSR_muon2017_bin13
34 prop_binSR_muon2018_bin3
35 prop_binSR_muon2017_bin15
36 prop_binSR_muon2018_bin12
37 pdf_Tot
38 prop_binCROS_muon2016M_bin2
39 prop_binCROS_electron2017_bin1
40 prop_binSR_electron2018_bin14
41 prop_binSR_electron2016M_bin0
42 OSest_electron_2018
1.00 +0.28 −0.28
43 prop_binSR_muon2018_bin8
44 prop_binSR_electron2018_bin17
45 prop_binSR_muon2017_bin1
46 OSest_electron_2017
1.00 +0.26 −0.26
47 prop_binCROS_electron2018_bin2
48 prop_binSR_electron2018_bin11
49 prop_binSR_electron2018_bin13
50 prop_binSR_muon2017_bin11
51 prop_binCRTT_muon2018_bin0
52 prop_binSR_muon2016M_bin12
53 prop_binSR_muon2016M_bin14
54 prop_binSR_electron2017_bin14
55 prop_binCROS_muon2017_bin0
56 QCDScale_WrongSign
57 prop_binCROS_muon2018_bin5
58 prop_binSR_muon2017_bin12
59 prop_binSR_muon2017_bin7
60 prop_binSR_electron2018_bin8
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂

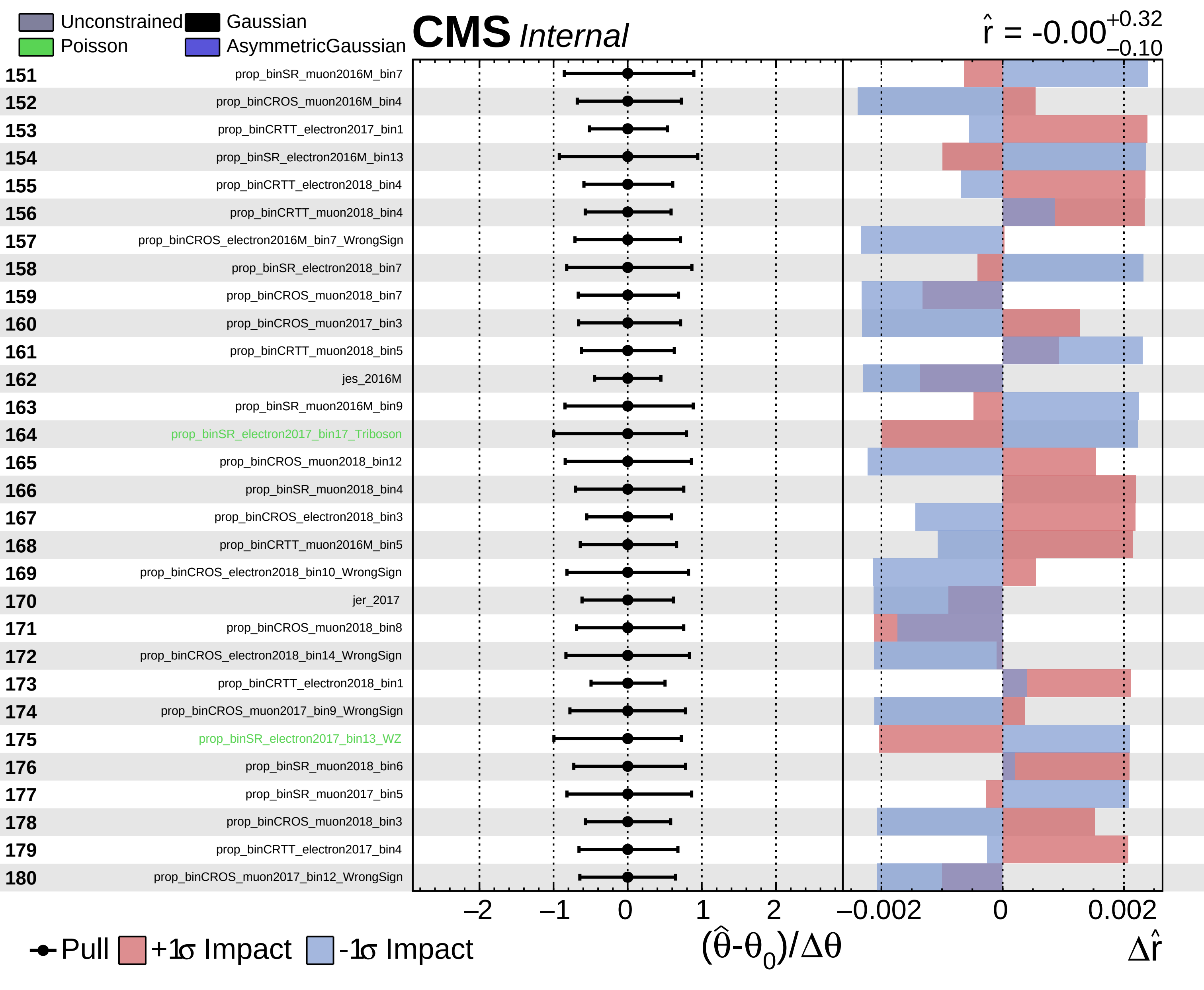

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = -0.00 +0.32 −0.10
61 prop_binCROS_electron2018_bin7
62 prop_binCRTT_muon2016M_bin0
63 prop_binSR_muon2017_bin16_ZZtoLep
64 prop_binSR_muon2018_bin7
65 prop_binSR_electron2018_bin2
66 prop_binSR_electron2017_bin12
67 prop_binSR_electron2017_bin8
68 prop_binCROS_electron2017_bin0
69 OSest_electron_2016M 1.00 +0.27 −0.27
70 OSest_muon_2017 1.00 +0.22 −0.22
71 prop_binSR_muon2016M_bin17
72 OSest_muon_2016M 1.00 +0.28 −0.28
73 prop_binSR_electron2017_bin10
74 prop_binSR_muon2017_bin6
75 prop_binSR_muon2016M_bin11
76 prop_binSR_electron2018_bin12
77 prop_binSR_muon2016M_bin15
78 prop_binSR_muon2016M_bin10
79 jer_2018
80 prop_binSR_electron2016M_bin10
81 prop_binSR_electron2018_bin9
82 prop_binSR_electron2017_bin1
83 prop_binSR_muon2016M_bin8
84 prop_binSR_muon2016M_bin13
85 prop_binSR_electron2017_bin5
86 ISR
87 prop_binSR_electron2017_bin11
88 QCDScale_TVX
89 prop_binSR_electron2016M_bin8
90 prop_binCROS_muon2017_bin6
−2 −1 0 1 2 −0.005 0 0.005
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δr̂



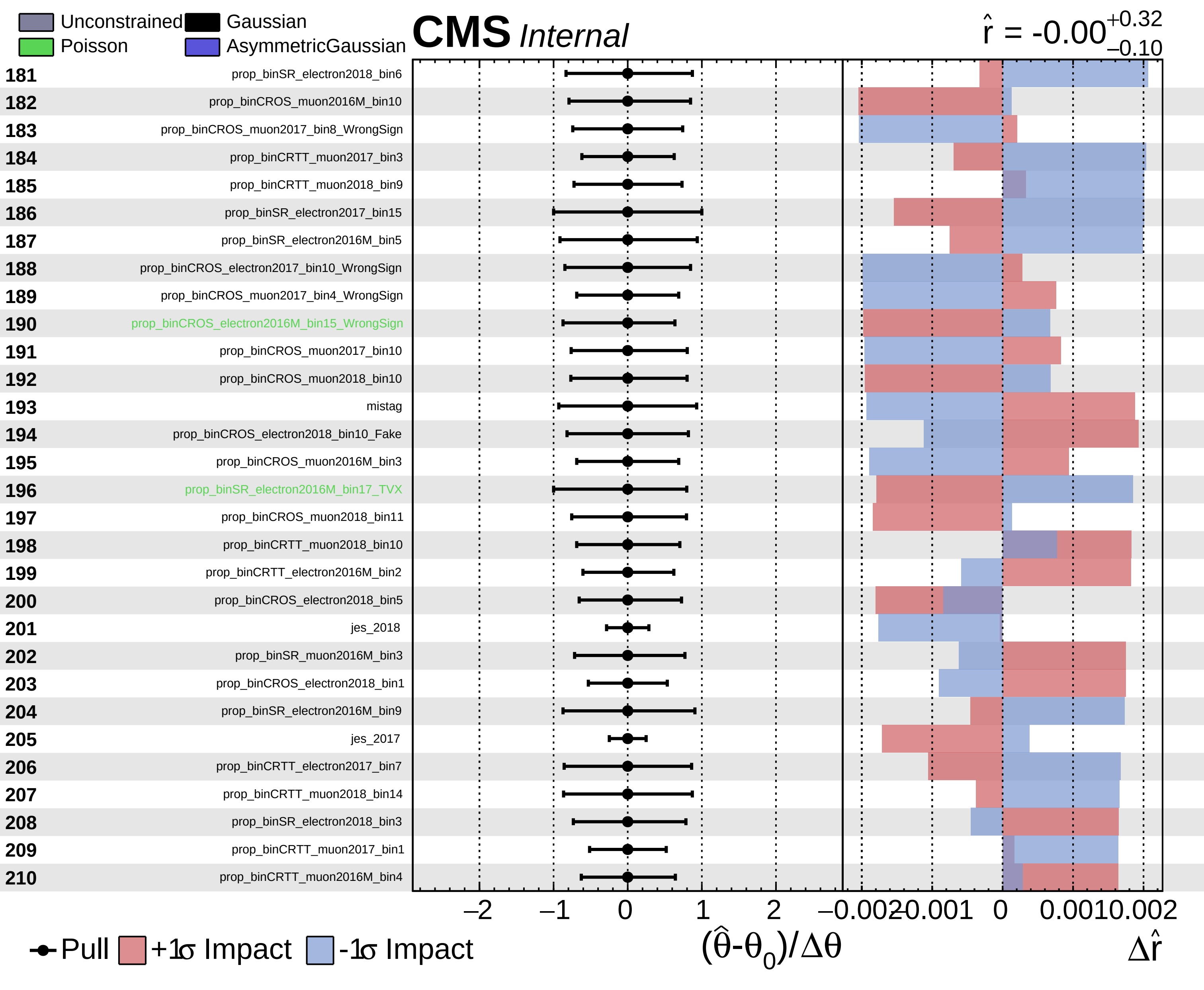
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.32}_{-0.10}$
91 prop_binCRTT_muon2018_bin3
92 prop_binCROS_muon2018_bin0
93 prop_binSR_muon2016M_bin6
94 prop_binSR_electron2016M_bin11
95 prop_binCROS_electron2016M_bin0
96 prop_binCROS_muon2016M_bin1
97 prop_binCRTT_muon2018_bin6
98 prop_binSR_electron2018_bin1
99 prop_binSR_muon2017_bin2
100 prop_binCROS_electron2018_bin15_WrongSign
101 prop_binSR_muon2017_bin10
102 btag
103 prop_binCROS_muon2018_bin6_WrongSign
104 prop_binSR_electron2016M_bin6
105 prop_binSR_electron2017_bin16
106 prop_binSR_electron2016M_bin16_WZ
107 prop_binSR_muon2017_bin8
108 prop_binCROS_muon2016M_bin9_WrongSign
109 prop_binSR_muon2016M_bin1
110 prop_binSR_electron2016M_bin12
111 prop_binCROS_muon2016M_bin0
112 prop_binSR_muon2017_bin16_TVX
113 prop_binCROS_electron2017_bin3
114 prop_binSR_electron2016M_bin15
115 prop_binSR_electron2018_bin10
116 prop_binSR_muon2016M_bin2
117 QCDScale_Triboson
118 prop_binCRTT_muon2018_bin2
119 prop_binSR_electron2016M_bin16_TTTo2L2Nu
120 prop_binSR_muon2017_bin9_WrongSign
−2 −1 0 1 2
−0.004 −0.002 0 0.002 0.004
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



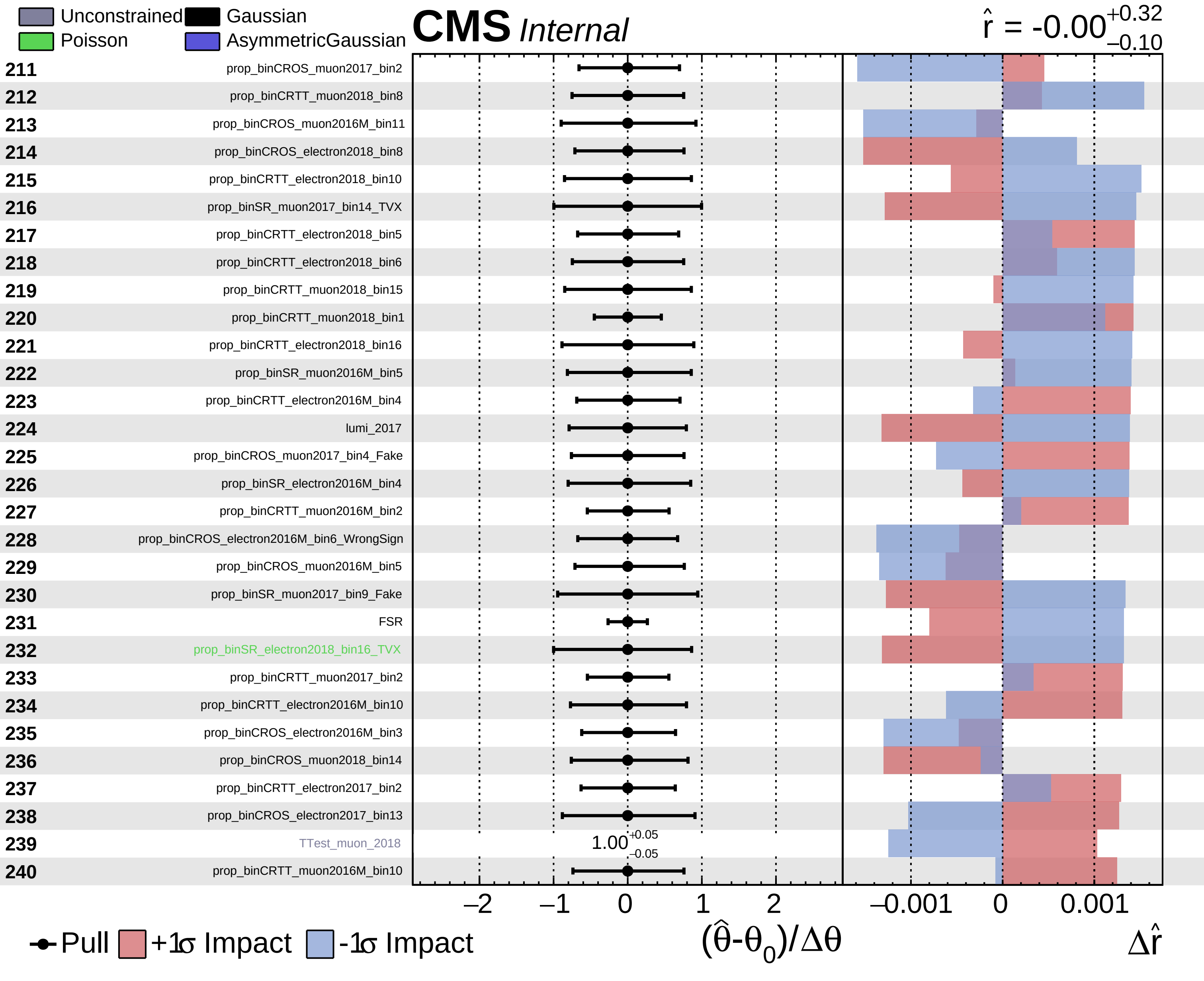
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.32}_{-0.10}$
121 prop_binCRTT_muon2017_bin0
122 prop_binCROS_electron2017_bin7
123 prop_binCRTT_electron2018_bin0
124 prop_binCRTT_muon2018_bin7
125 prop_binSR_electron2016M_bin14
126 prop_binCROS_muon2018_bin2
127 TES
128 prop_binCRTT_electron2017_bin0
129 prop_binCRTT_muon2016M_bin1
130 prop_binSR_electron2017_bin7
131 prop_binCRTT_electron2016M_bin0
132 prop_binCROS_electron2016M_bin1
133 prop_binSR_electron2017_bin13_Fake
134 prop_binCROS_muon2018_bin9
135 prop_binCROS_muon2017_bin1
136 PF
137 prop_binCRTT_muon2018_bin13
138 prop_binSR_electron2016M_bin1
139 prop_binCRTT_muon2018_bin11
140 prop_binCROS_electron2017_bin4
141 prop_binSR_muon2018_bin5
142 prop_binCROS_electron2018_bin6
143 prop_binCROS_muon2018_bin1
144 prop_binCRTT_electron2018_bin3
145 prop_binSR_muon2017_bin17
146 prop_binSR_electron2017_bin9
147 prop_binSR_electron2017_bin6
148 prop_binCRTT_electron2016M_bin5
149 prop_binCROS_electron2017_bin9_WrongSign
150 prop_binCRTT_electron2018_bin2
−2
−1
0
1
2
−0.002
0
0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

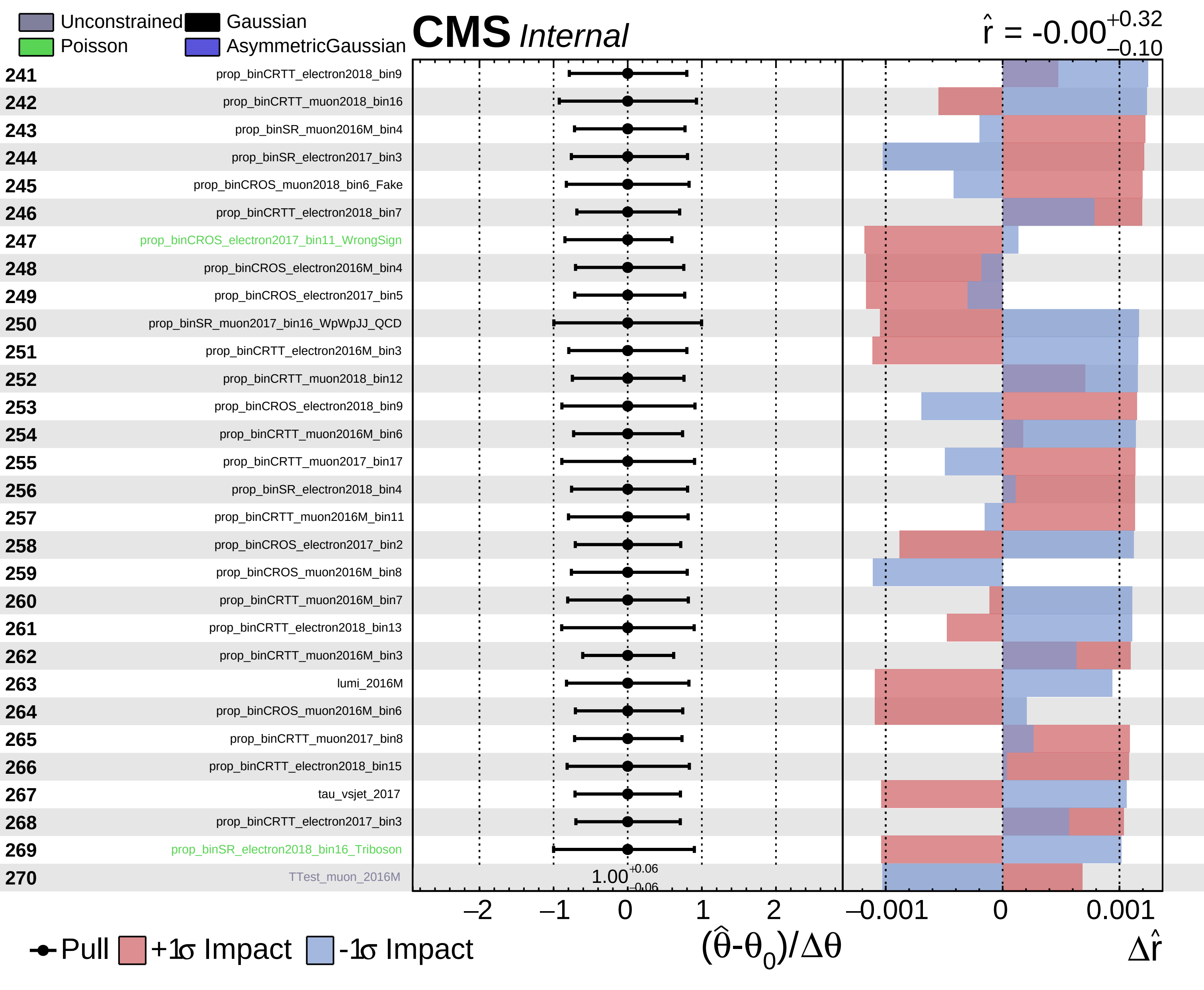


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.32}_{-0.10}$
151 prop_binSR_muon2016M_bin7
152 prop_binCROS_muon2016M_bin4
153 prop_binCRTT_electron2017_bin1
154 prop_binSR_electron2016M_bin13
155 prop_binCRTT_electron2018_bin4
156 prop_binCRTT_muon2018_bin4
157 prop_binCROS_electron2016M_bin7_WrongSign
158 prop_binSR_electron2018_bin7
159 prop_binCROS_muon2018_bin7
160 prop_binCROS_muon2017_bin3
161 prop_binCRTT_muon2018_bin5
162 jes_2016M
163 prop_binSR_muon2016M_bin9
164 prop_binSR_electron2017_bin17_Triboson
165 prop_binCROS_muon2018_bin12
166 prop_binSR_muon2018_bin4
167 prop_binCROS_electron2018_bin3
168 prop_binCRTT_muon2016M_bin5
169 prop_binCROS_electron2018_bin10_WrongSign
170 jer_2017
171 prop_binCROS_muon2018_bin8
172 prop_binCROS_electron2018_bin14_WrongSign
173 prop_binCRTT_electron2018_bin1
174 prop_binCROS_muon2017_bin9_WrongSign
175 prop_binSR_electron2017_bin13_WZ
176 prop_binSR_muon2018_bin6
177 prop_binSR_muon2017_bin5
178 prop_binCROS_muon2018_bin3
179 prop_binCRTT_electron2017_bin4
180 prop_binCROS_muon2017_bin12_WrongSign
−2
−1
0
1
2
−0.002
0
0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

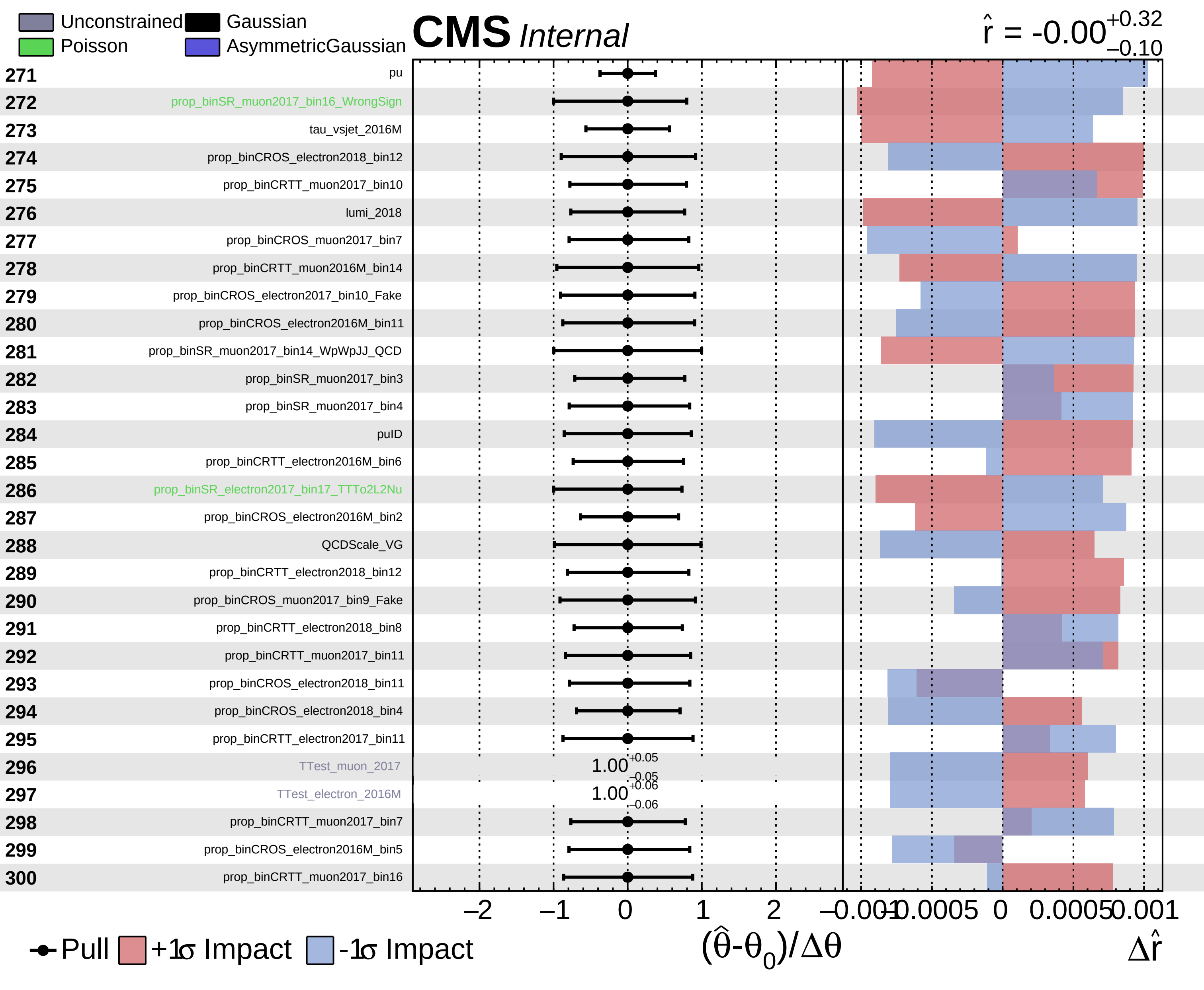


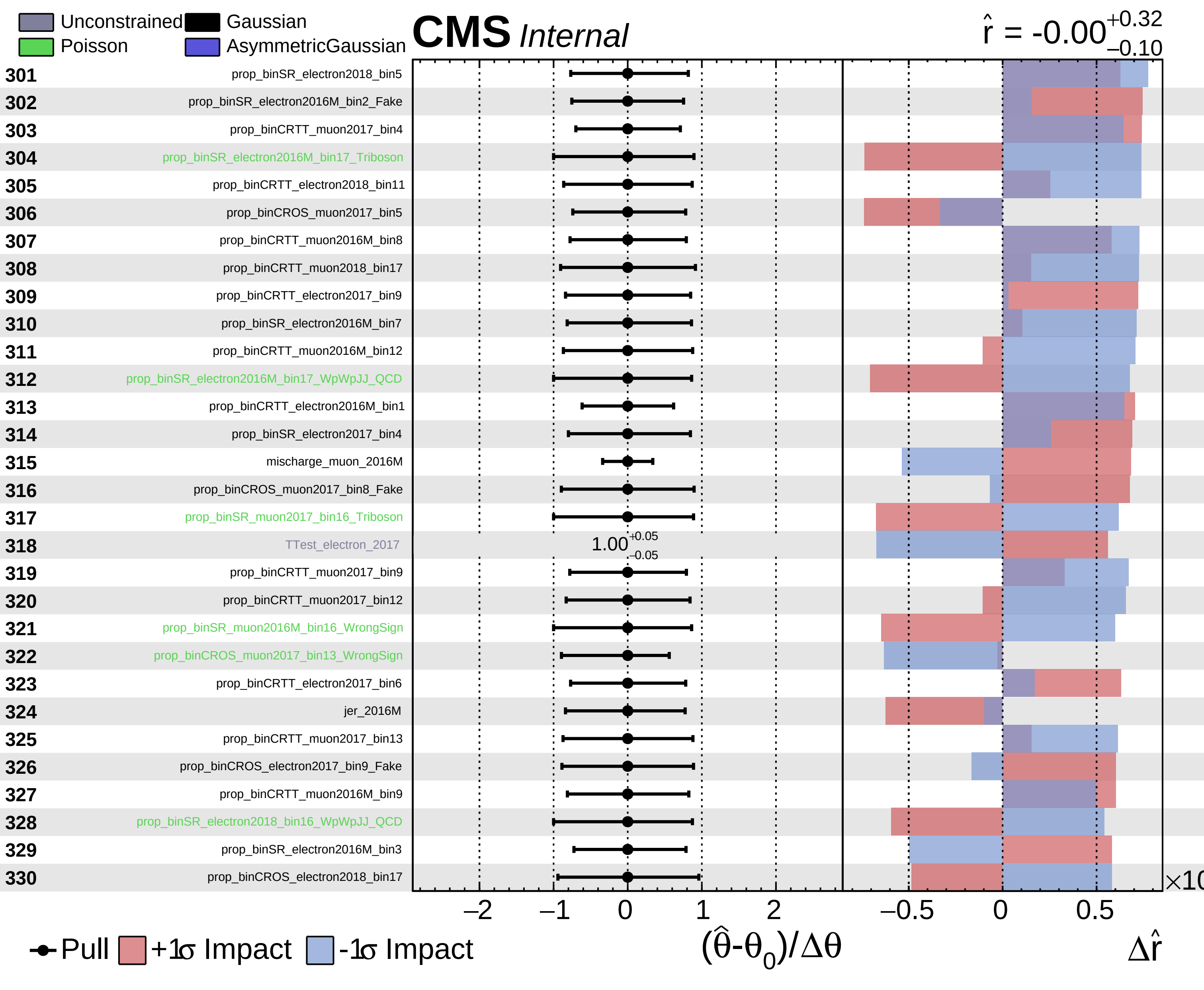
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.32}_{-0.10}$
181 prop_binSR_electron2018_bin6
182 prop_binCROS_muon2016M_bin10
183 prop_binCROS_muon2017_bin8_WrongSign
184 prop_binCRTT_muon2017_bin3
185 prop_binCRTT_muon2018_bin9
186 prop_binSR_electron2017_bin15
187 prop_binSR_electron2016M_bin5
188 prop_binCROS_electron2017_bin10_WrongSign
189 prop_binCROS_muon2017_bin4_WrongSign
190 prop_binCROS_electron2016M_bin15_WrongSign
191 prop_binCROS_muon2017_bin10
192 prop_binCROS_muon2018_bin10
193 mistag
194 prop_binCROS_electron2018_bin10_Fake
195 prop_binCROS_muon2016M_bin3
196 prop_binSR_electron2016M_bin17_TVX
197 prop_binCROS_muon2018_bin11
198 prop_binCRTT_muon2018_bin10
199 prop_binCRTT_electron2016M_bin2
200 prop_binCROS_electron2018_bin5
201 jes_2018
202 prop_binSR_muon2016M_bin3
203 prop_binCROS_electron2018_bin1
204 prop_binSR_electron2016M_bin9
205 jes_2017
206 prop_binCRTT_electron2017_bin7
207 prop_binCRTT_muon2018_bin14
208 prop_binSR_electron2018_bin3
209 prop_binCRTT_muon2017_bin1
210 prop_binCRTT_muon2016M_bin4
−2 −1 0 1 2
−0.002 −0.001 0 0.001 0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.32}_{-0.10}$
211 prop_binCROS_muon2017_bin2
212 prop_binCRTT_muon2018_bin8
213 prop_binCROS_muon2016M_bin11
214 prop_binCROS_electron2018_bin8
215 prop_binCRTT_electron2018_bin10
216 prop_binSR_muon2017_bin14_TVX
217 prop_binCRTT_electron2018_bin5
218 prop_binCRTT_electron2018_bin6
219 prop_binCRTT_muon2018_bin15
220 prop_binCRTT_muon2018_bin1
221 prop_binCRTT_electron2018_bin16
222 prop_binSR_muon2016M_bin5
223 prop_binCRTT_electron2016M_bin4
224 lumi_2017
225 prop_binCROS_muon2017_bin4_Fake
226 prop_binSR_electron2016M_bin4
227 prop_binCRTT_muon2016M_bin2
228 prop_binCROS_electron2016M_bin6_WrongSign
229 prop_binCROS_muon2016M_bin5
230 prop_binSR_muon2017_bin9_Fake
231 FSR
232 prop_binSR_electron2018_bin16_TVX
233 prop_binCRTT_muon2017_bin2
234 prop_binCRTT_electron2016M_bin10
235 prop_binCROS_electron2016M_bin3
236 prop_binCROS_muon2018_bin14
237 prop_binCRTT_electron2017_bin2
238 prop_binCROS_electron2017_bin13
239 TTest_muon_2018
$1.00^{+0.05}_{-0.05}$
240 prop_binCRTT_muon2016M_bin10
−2
−1
0
1
2
−0.001
0
0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

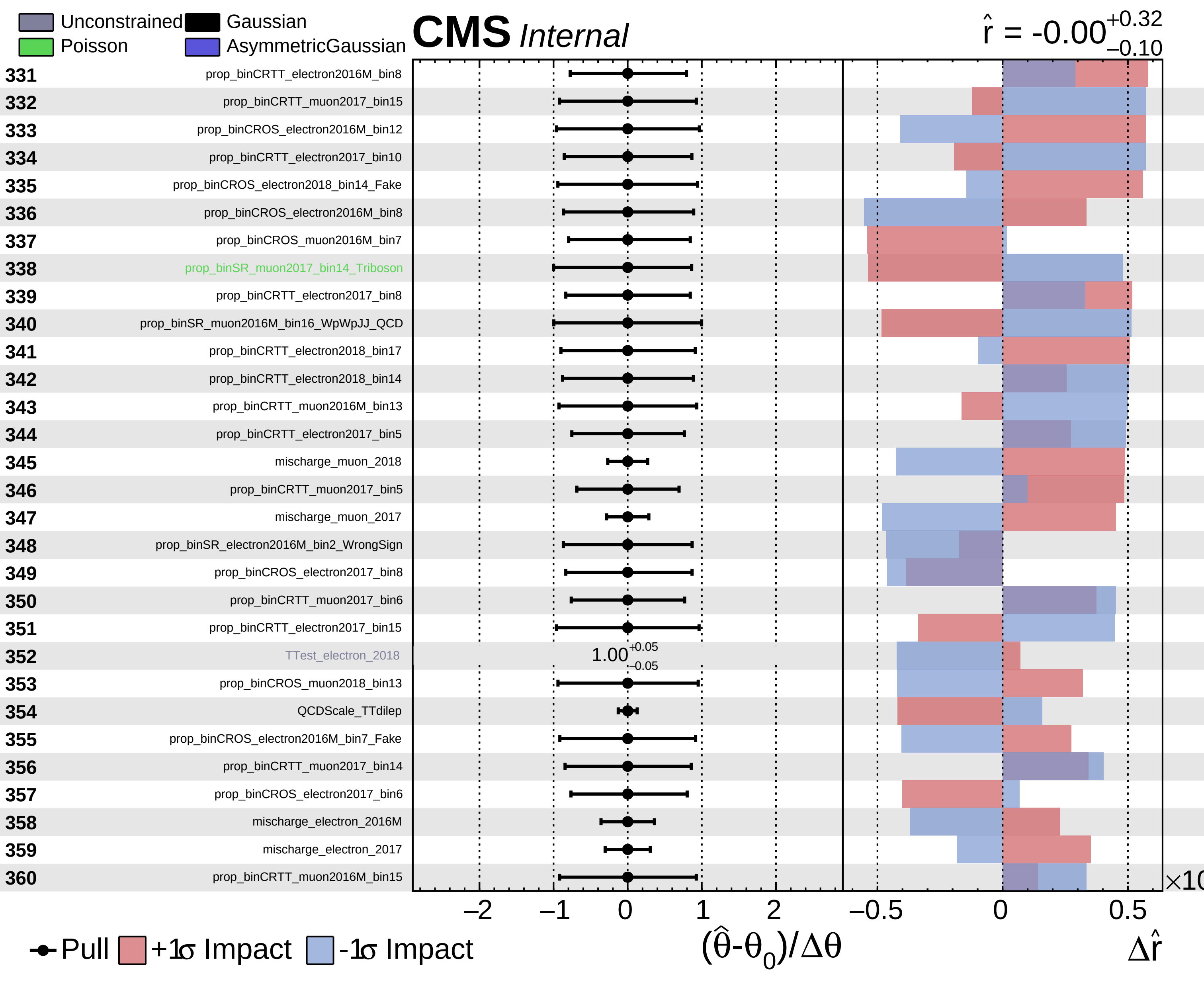


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.32}_{-0.10}$
241 prop_binCRTT_electron2018_bin9
242 prop_binCRTT_muon2018_bin16
243 prop_binSR_muon2016M_bin4
244 prop_binSR_electron2017_bin3
245 prop_binCROS_muon2018_bin6_Fake
246 prop_binCRTT_electron2018_bin7
247 prop_binCROS_electron2017_bin11_WrongSign
248 prop_binCROS_electron2016M_bin4
249 prop_binCROS_electron2017_bin5
250 prop_binSR_muon2017_bin16_WpWpJJ_QCD
251 prop_binCRTT_electron2016M_bin3
252 prop_binCRTT_muon2018_bin12
253 prop_binCROS_electron2018_bin9
254 prop_binCRTT_muon2016M_bin6
255 prop_binCRTT_muon2017_bin17
256 prop_binSR_electron2018_bin4
257 prop_binCRTT_muon2016M_bin11
258 prop_binCROS_electron2017_bin2
259 prop_binCROS_muon2016M_bin8
260 prop_binCRTT_muon2016M_bin7
261 prop_binCRTT_electron2018_bin13
262 prop_binCRTT_muon2016M_bin3
263 lumi_2016M
264 prop_binCROS_muon2016M_bin6
265 prop_binCRTT_muon2017_bin8
266 prop_binCRTT_electron2018_bin15
267 tau_vsjet_2017
268 prop_binCRTT_electron2017_bin3
269 prop_binSR_electron2018_bin16_Triboson
270 TTest_muon_2016M
$1.00^{+0.06}_{-0.06}$
−2
−1
0
1
2
−0.001
0
0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

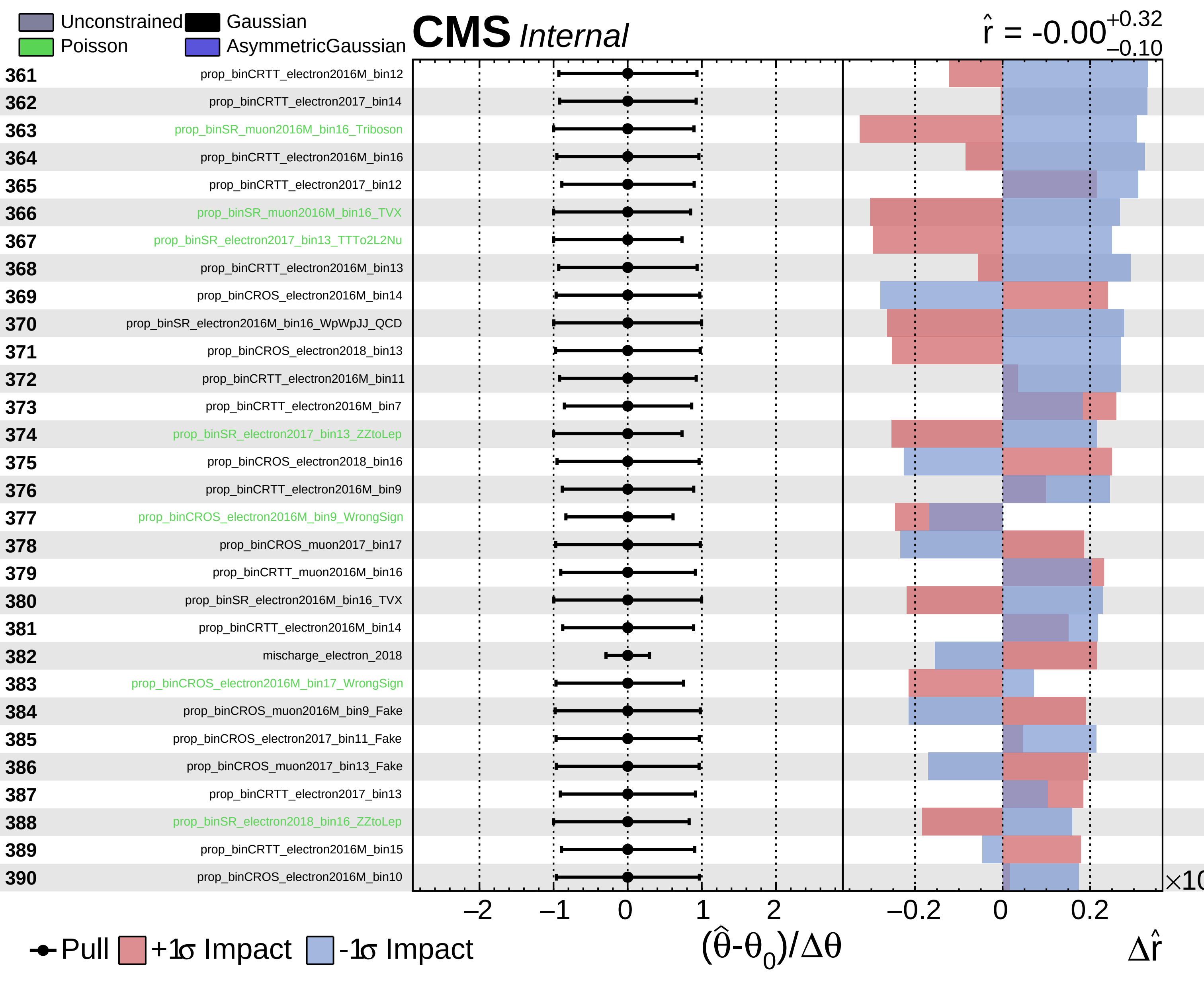

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.32}_{-0.10}$
271 pu
272 prop_binSR_muon2017_bin16_WrongSign
273 tau_vsjet_2016M
274 prop_binCROS_electron2018_bin12
275 prop_binCRTT_muon2017_bin10
276 lumi_2018
277 prop_binCROS_muon2017_bin7
278 prop_binCRTT_muon2016M_bin14
279 prop_binCROS_electron2017_bin10_Fake
280 prop_binCROS_electron2016M_bin11
281 prop_binSR_muon2017_bin14_WpWpJJ_QCD
282 prop_binSR_muon2017_bin3
283 prop_binSR_muon2017_bin4
284 puID
285 prop_binCRTT_electron2016M_bin6
286 prop_binSR_electron2017_bin17_TTTo2L2Nu
287 prop_binCROS_electron2016M_bin2
288 QCDScale_VG
289 prop_binCRTT_electron2018_bin12
290 prop_binCROS_muon2017_bin9_Fake
291 prop_binCRTT_electron2018_bin8
292 prop_binCRTT_muon2017_bin11
293 prop_binCROS_electron2018_bin11
294 prop_binCROS_electron2018_bin4
295 prop_binCRTT_electron2017_bin11
296 TTest_muon_2017 $1.00^{+0.05}_{-0.05}$
297 TTest_electron_2016M $1.00^{+0.06}_{-0.06}$
298 prop_binCRTT_muon2017_bin7
299 prop_binCROS_electron2016M_bin5
300 prop_binCRTT_muon2017_bin16
−2 −1 0 1 2
−0.001 −0.0005 0 0.0005 0.001
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



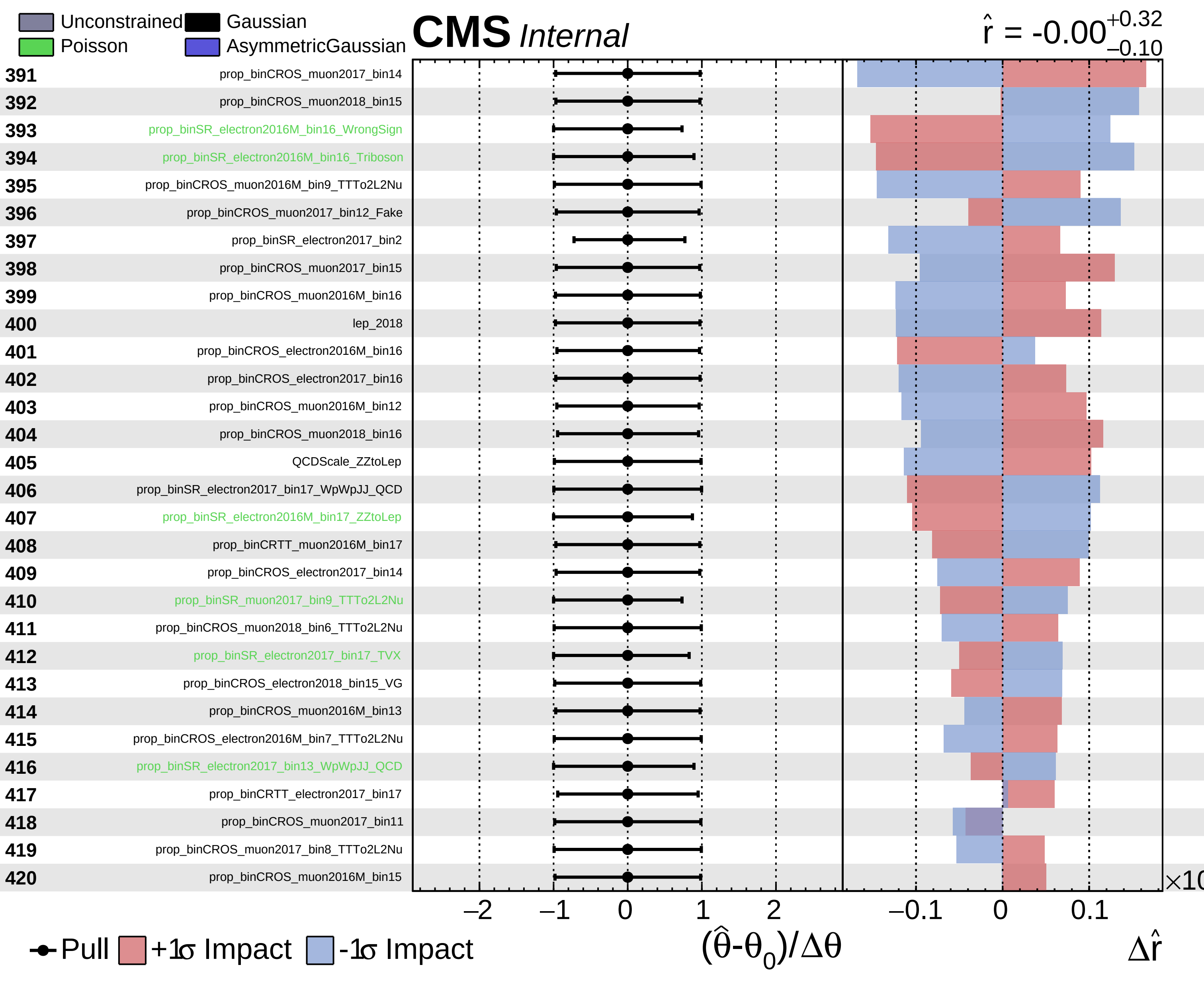
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.32}_{-0.10}$
301 prop_binSR_electron2018_bin5
302 prop_binSR_electron2016M_bin2_Fake
303 prop_binCRTT_muon2017_bin4
304 prop_binSR_electron2016M_bin17_Triboson
305 prop_binCRTT_electron2018_bin11
306 prop_binCROS_muon2017_bin5
307 prop_binCRTT_muon2016M_bin8
308 prop_binCRTT_muon2018_bin17
309 prop_binCRTT_electron2017_bin9
310 prop_binSR_electron2016M_bin7
311 prop_binCRTT_muon2016M_bin12
312 prop_binSR_electron2016M_bin17_WpWpJJ_QCD
313 prop_binCRTT_electron2016M_bin1
314 prop_binSR_electron2017_bin4
315 mischarge_muon_2016M
316 prop_binCROS_muon2017_bin8_Fake
317 prop_binSR_muon2017_bin16_Triboson
318 TTest_electron_2017
$1.00^{+0.05}_{-0.05}$
319 prop_binCRTT_muon2017_bin9
320 prop_binCRTT_muon2017_bin12
321 prop_binSR_muon2016M_bin16_WrongSign
322 prop_binCROS_muon2017_bin13_WrongSign
323 prop_binCRTT_electron2017_bin6
324 jer_2016M
325 prop_binCRTT_muon2017_bin13
326 prop_binCROS_electron2017_bin9_Fake
327 prop_binCRTT_muon2016M_bin9
328 prop_binSR_electron2018_bin16_WpWpJJ_QCD
329 prop_binSR_electron2016M_bin3
330 prop_binCROS_electron2018_bin17
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

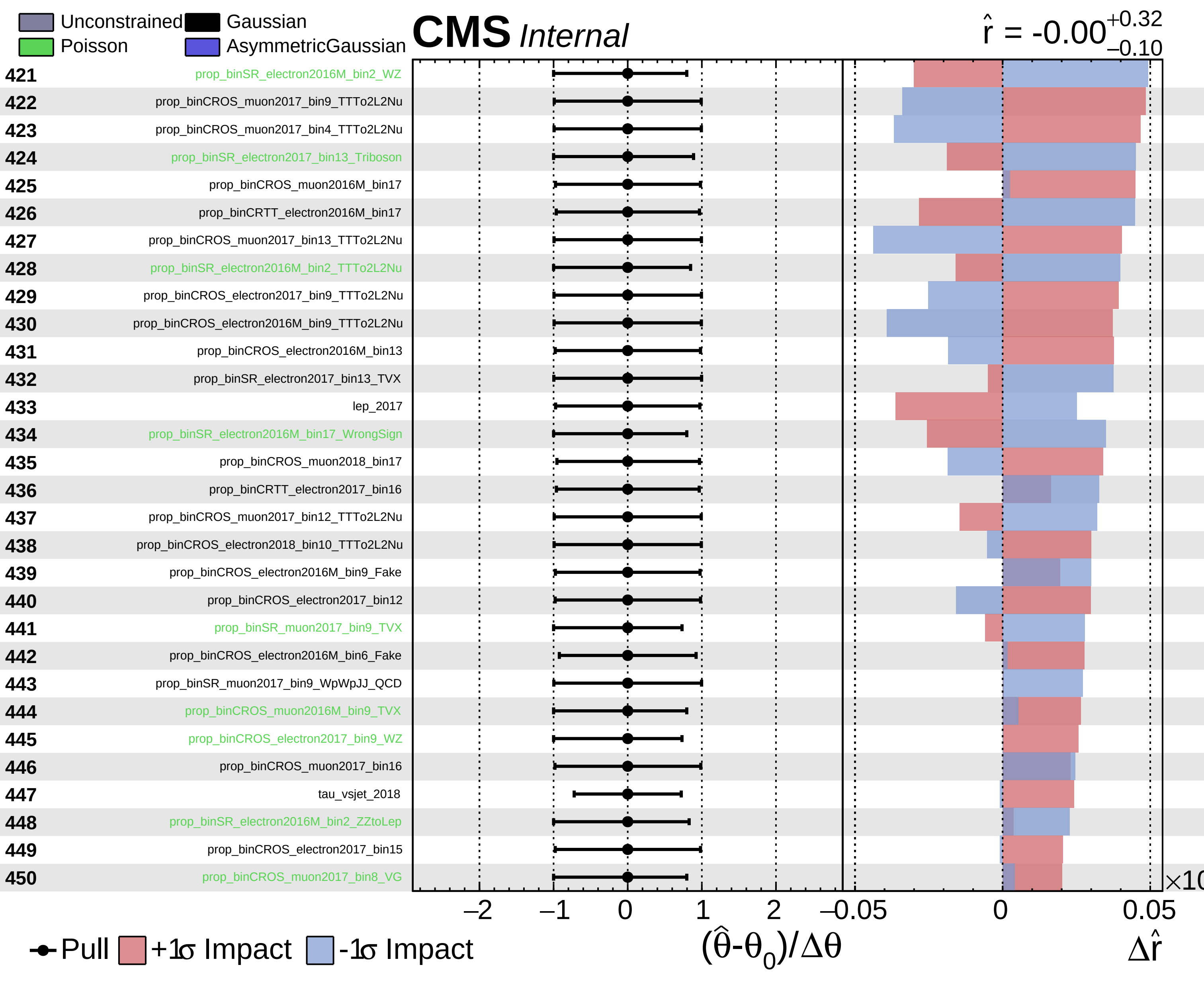


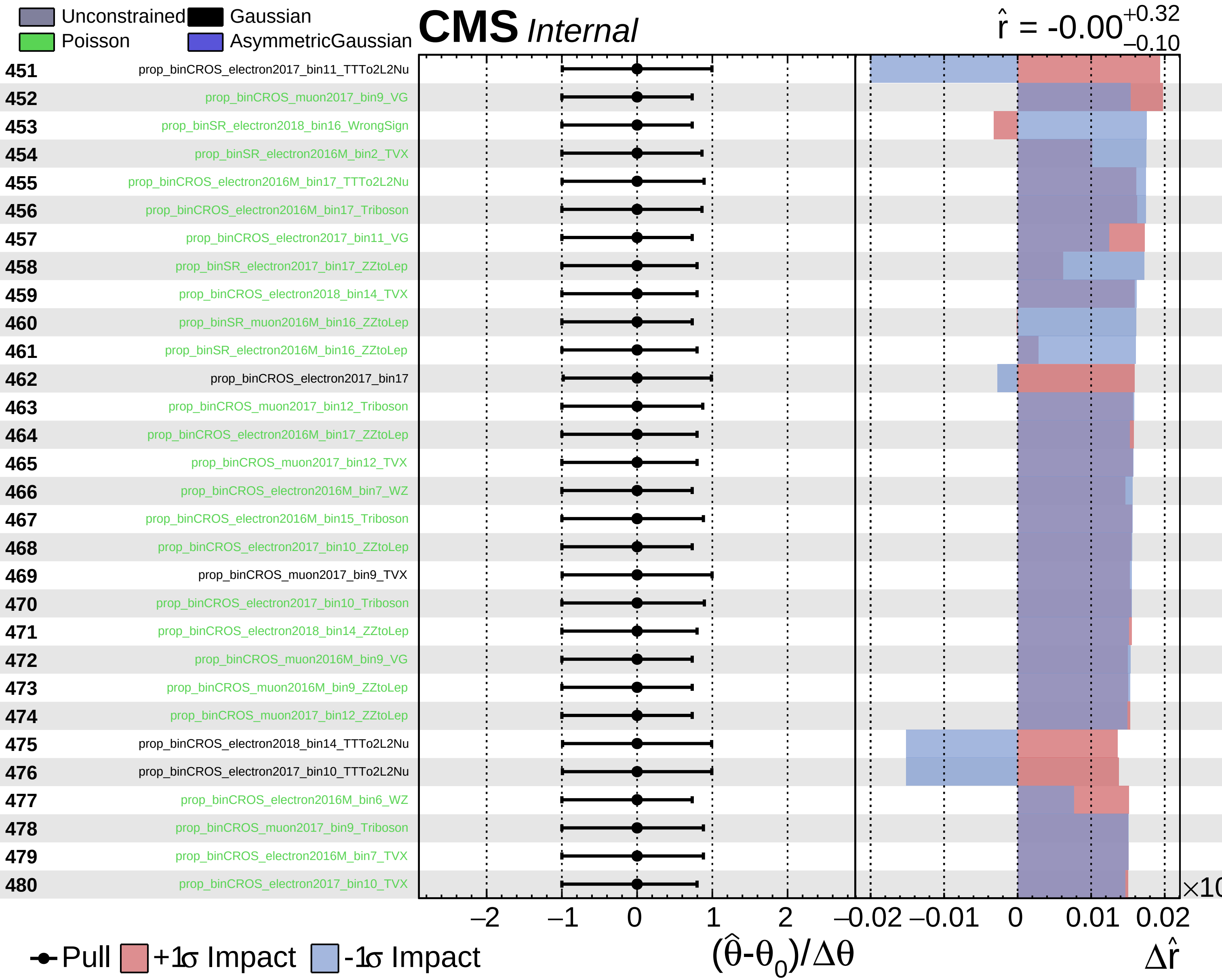
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.32}_{-0.10}$
331 prop_binCRTT_electron2016M_bin8
332 prop_binCRTT_muon2017_bin15
333 prop_binCROS_electron2016M_bin12
334 prop_binCRTT_electron2017_bin10
335 prop_binCROS_electron2018_bin14_Fake
336 prop_binCROS_electron2016M_bin8
337 prop_binCROS_muon2016M_bin7
338 prop_binSR_muon2017_bin14_Triboson
339 prop_binCRTT_electron2017_bin8
340 prop_binSR_muon2016M_bin16_WpWpJJ_QCD
341 prop_binCRTT_electron2018_bin17
342 prop_binCRTT_electron2018_bin14
343 prop_binCRTT_muon2016M_bin13
344 prop_binCRTT_electron2017_bin5
345 mischarge_muon_2018
346 prop_binCRTT_muon2017_bin5
347 mischarge_muon_2017
348 prop_binSR_electron2016M_bin2_WrongSign
349 prop_binCROS_electron2017_bin8
350 prop_binCRTT_muon2017_bin6
351 prop_binCRTT_electron2017_bin15
352 TTest_electron_2018
$1.00^{+0.05}_{-0.05}$
353 prop_binCROS_muon2018_bin13
354 QCDScale_TTdilep
355 prop_binCROS_electron2016M_bin7_Fake
356 prop_binCRTT_muon2017_bin14
357 prop_binCROS_electron2017_bin6
358 mischarge_electron_2016M
359 mischarge_electron_2017
360 prop_binCRTT_muon2016M_bin15
−2
−1
0
1
2
−0.5
0
0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

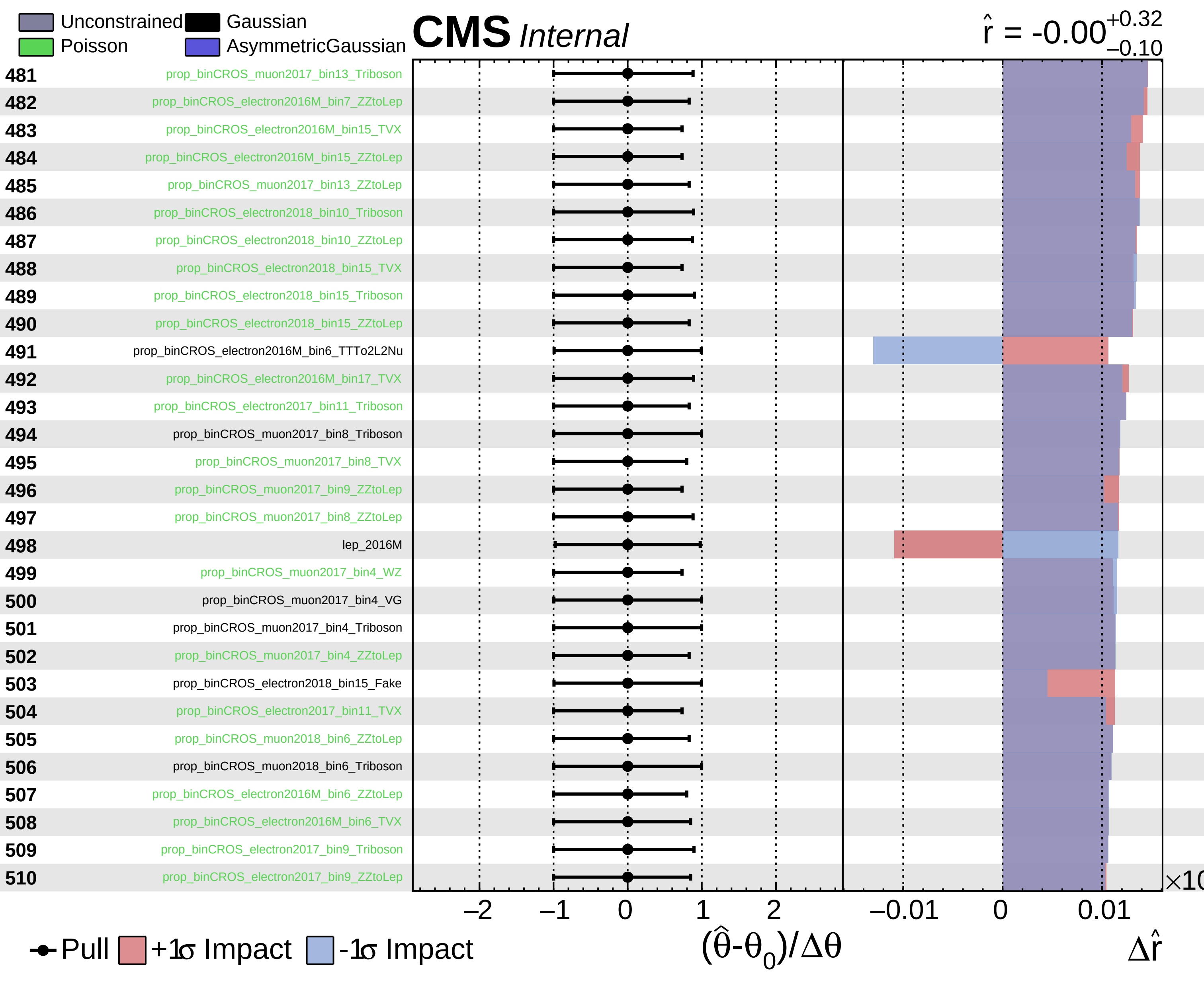


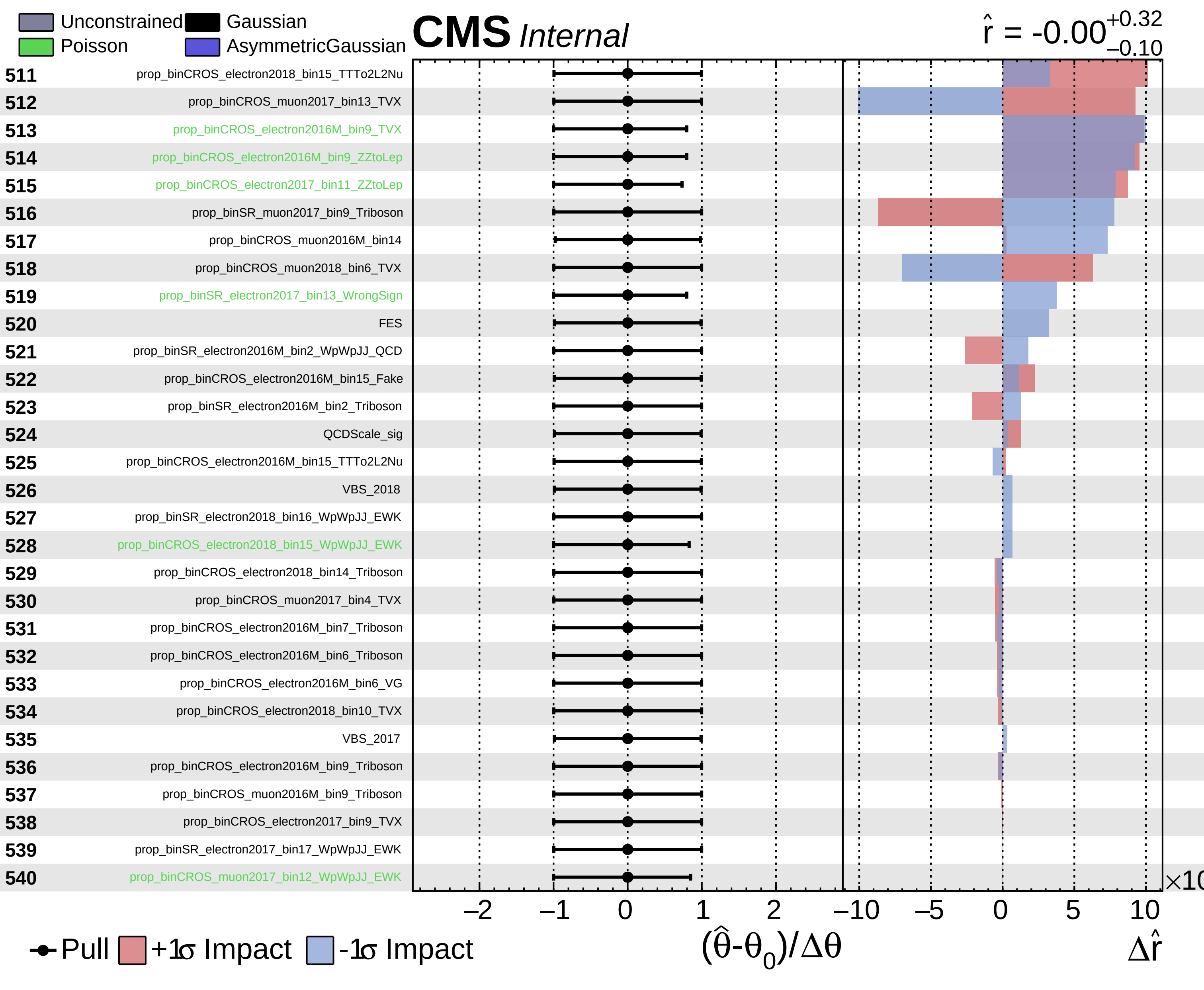
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.32}_{-0.10}$
361 prop_binCRTT_electron2016M_bin12
362 prop_binCRTT_electron2017_bin14
363 prop_binSR_muon2016M_bin16_Triboson
364 prop_binCRTT_electron2016M_bin16
365 prop_binCRTT_electron2017_bin12
366 prop_binSR_muon2016M_bin16_TVX
367 prop_binSR_electron2017_bin13_TTTo2L2Nu
368 prop_binCRTT_electron2016M_bin13
369 prop_binCROS_electron2016M_bin14
370 prop_binSR_electron2016M_bin16_WpWpJJ_QCD
371 prop_binCROS_electron2018_bin13
372 prop_binCRTT_electron2016M_bin11
373 prop_binCRTT_electron2016M_bin7
374 prop_binSR_electron2017_bin13_ZZtoLep
375 prop_binCROS_electron2018_bin16
376 prop_binCRTT_electron2016M_bin9
377 prop_binCROS_electron2016M_bin9_WrongSign
378 prop_binCROS_muon2017_bin17
379 prop_binCRTT_muon2016M_bin16
380 prop_binSR_electron2016M_bin16_TVX
381 prop_binCRTT_electron2016M_bin14
382 mischarge_electron_2018
383 prop_binCROS_electron2016M_bin17_WrongSign
384 prop_binCROS_muon2016M_bin9_Fake
385 prop_binCROS_electron2017_bin11_Fake
386 prop_binCROS_muon2017_bin13_Fake
387 prop_binCRTT_electron2017_bin13
388 prop_binSR_electron2018_bin16_ZZtoLep
389 prop_binCRTT_electron2016M_bin15
390 prop_binCROS_electron2016M_bin10
−2 −1 0 1 2
−0.2 0 0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.32}_{-0.10}$
391 prop_binCROS_muon2017_bin14
392 prop_binCROS_muon2018_bin15
393 prop_binSR_electron2016M_bin16_WrongSign
394 prop_binSR_electron2016M_bin16_Triboson
395 prop_binCROS_muon2016M_bin9_TTTo2L2Nu
396 prop_binCROS_muon2017_bin12_Fake
397 prop_binSR_electron2017_bin2
398 prop_binCROS_muon2017_bin15
399 prop_binCROS_muon2016M_bin16
400 lep_2018
401 prop_binCROS_electron2016M_bin16
402 prop_binCROS_electron2017_bin16
403 prop_binCROS_muon2016M_bin12
404 prop_binCROS_muon2018_bin16
405 QCDScale_ZZtoLep
406 prop_binSR_electron2017_bin17_WpWpJJ_QCD
407 prop_binSR_electron2016M_bin17_ZZtoLep
408 prop_binCRTT_muon2016M_bin17
409 prop_binCROS_electron2017_bin14
410 prop_binSR_muon2017_bin9_TTTo2L2Nu
411 prop_binCROS_muon2018_bin6_TTTo2L2Nu
412 prop_binSR_electron2017_bin17_TVX
413 prop_binCROS_electron2018_bin15_VG
414 prop_binCROS_muon2016M_bin13
415 prop_binCROS_electron2016M_bin7_TTTo2L2Nu
416 prop_binSR_electron2017_bin13_WpWpJJ_QCD
417 prop_binCRTT_electron2017_bin17
418 prop_binCROS_muon2017_bin11
419 prop_binCROS_muon2017_bin8_TTTo2L2Nu
420 prop_binCROS_muon2016M_bin15
−2 −1 0 1 2
−0.1 0 0.1
×10
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

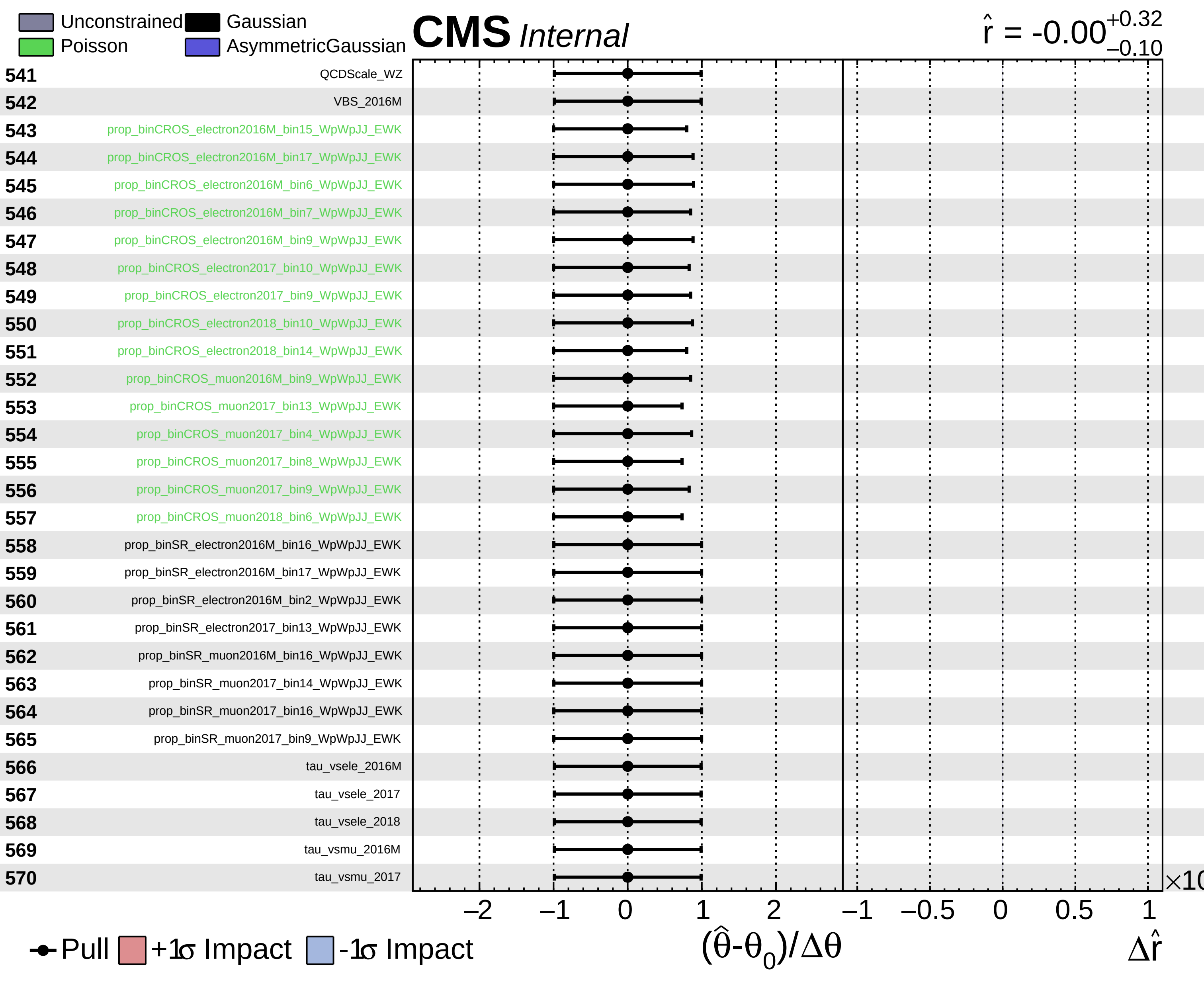

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.32}_{-0.10}$
421 prop_binSR_electron2016M_bin2_WZ
422 prop_binCROS_muon2017_bin9_TTTo2L2Nu
423 prop_binCROS_muon2017_bin4_TTTo2L2Nu
424 prop_binSR_electron2017_bin13_Triboson
425 prop_binCROS_muon2016M_bin17
426 prop_binCRTT_electron2016M_bin17
427 prop_binCROS_muon2017_bin13_TTTo2L2Nu
428 prop_binSR_electron2016M_bin2_TTTo2L2Nu
429 prop_binCROS_electron2017_bin9_TTTo2L2Nu
430 prop_binCROS_electron2016M_bin9_TTTo2L2Nu
431 prop_binCROS_electron2016M_bin13
432 prop_binSR_electron2017_bin13_TVX
433 lep_2017
434 prop_binSR_electron2016M_bin17_WrongSign
435 prop_binCROS_muon2018_bin17
436 prop_binCRTT_electron2017_bin16
437 prop_binCROS_muon2017_bin12_TTTo2L2Nu
438 prop_binCROS_electron2018_bin10_TTTo2L2Nu
439 prop_binCROS_electron2016M_bin9_Fake
440 prop_binCROS_electron2017_bin12
441 prop_binSR_muon2017_bin9_TVX
442 prop_binCROS_electron2016M_bin6_Fake
443 prop_binSR_muon2017_bin9_WpWpJJ_QCD
444 prop_binCROS_muon2016M_bin9_TVX
445 prop_binCROS_electron2017_bin9_WZ
446 prop_binCROS_muon2017_bin16
447 tau_vsjet_2018
448 prop_binSR_electron2016M_bin2_ZZtoLep
449 prop_binCROS_electron2017_bin15
450 prop_binCROS_muon2017_bin8_VG
−2
−1
0
1
2
−0.05
0
0.05
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

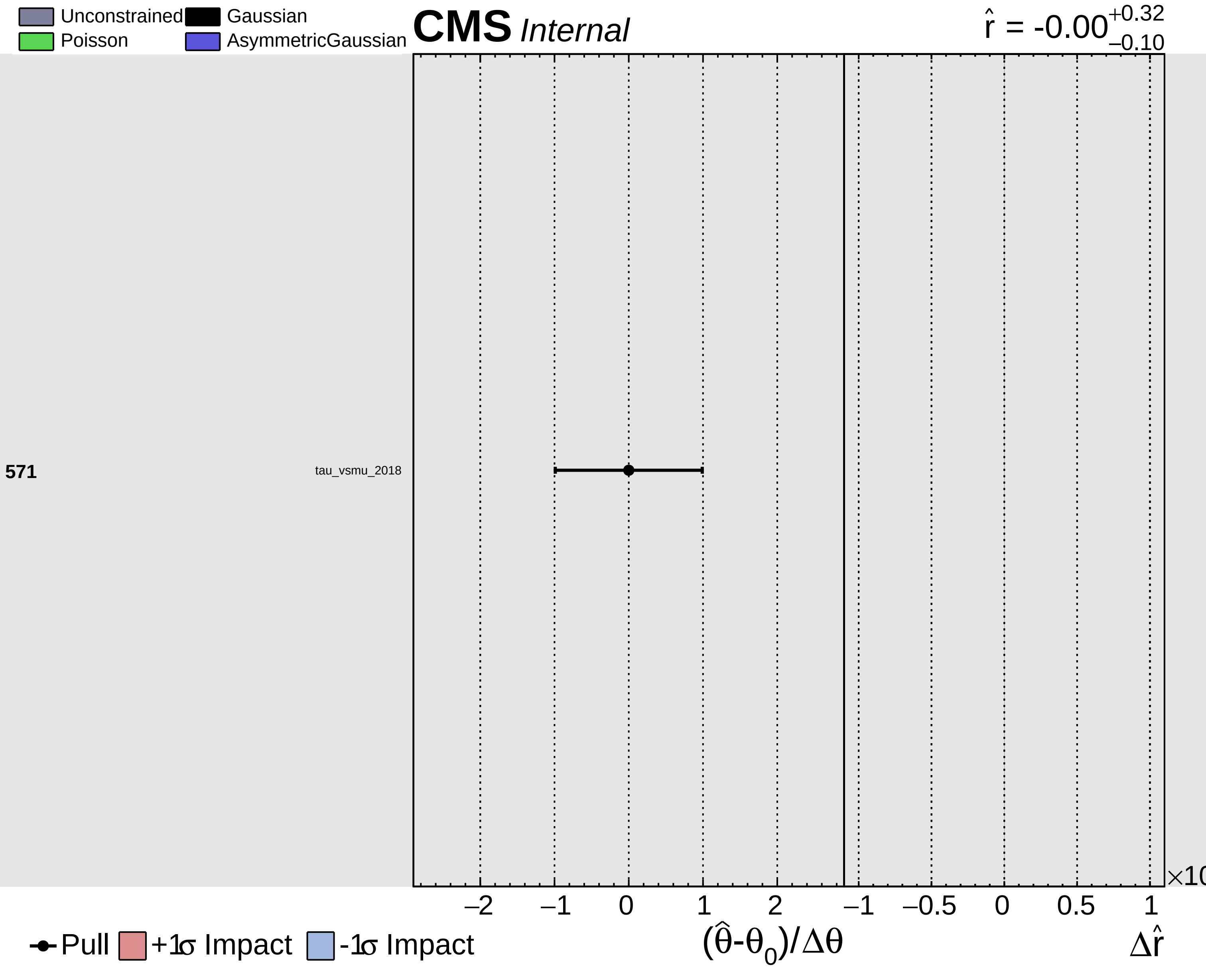

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.32}_{-0.10}$
451 prop_binCROS_electron2017_bin11_TTTo2L2Nu
452 prop_binCROS_muon2017_bin9_VG
453 prop_binSR_electron2018_bin16_WrongSign
454 prop_binSR_electron2016M_bin2_TVX
455 prop_binCROS_electron2016M_bin17_TTTo2L2Nu
456 prop_binCROS_electron2016M_bin17_Triboson
457 prop_binCROS_electron2017_bin11_VG
458 prop_binSR_electron2017_bin17_ZZtoLep
459 prop_binCROS_electron2018_bin14_TVX
460 prop_binSR_muon2016M_bin16_ZZtoLep
461 prop_binSR_electron2016M_bin16_ZZtoLep
462 prop_binCROS_electron2017_bin17
463 prop_binCROS_muon2017_bin12_Triboson
464 prop_binCROS_electron2016M_bin17_ZZtoLep
465 prop_binCROS_muon2017_bin12_TVX
466 prop_binCROS_electron2016M_bin7_WZ
467 prop_binCROS_electron2016M_bin15_Triboson
468 prop_binCROS_electron2017_bin10_ZZtoLep
469 prop_binCROS_muon2017_bin9_TVX
470 prop_binCROS_electron2017_bin10_Triboson
471 prop_binCROS_electron2018_bin14_ZZtoLep
472 prop_binCROS_muon2016M_bin9_VG
473 prop_binCROS_muon2016M_bin9_ZZtoLep
474 prop_binCROS_muon2017_bin12_ZZtoLep
475 prop_binCROS_electron2018_bin14_TTTo2L2Nu
476 prop_binCROS_electron2017_bin10_TTTo2L2Nu
477 prop_binCROS_electron2016M_bin6_WZ
478 prop_binCROS_muon2017_bin9_Triboson
479 prop_binCROS_electron2016M_bin7_TVX
480 prop_binCROS_electron2017_bin10_TVX
−2 −1 0 1 2
−0.02 −0.01 0 0.01 0.02
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.32}_{-0.10}$
481 prop_binCROS_muon2017_bin13_Triboson
482 prop_binCROS_electron2016M_bin7_ZZtoLep
483 prop_binCROS_electron2016M_bin15_TVX
484 prop_binCROS_electron2016M_bin15_ZZtoLep
485 prop_binCROS_muon2017_bin13_ZZtoLep
486 prop_binCROS_electron2018_bin10_Triboson
487 prop_binCROS_electron2018_bin10_ZZtoLep
488 prop_binCROS_electron2018_bin15_TVX
489 prop_binCROS_electron2018_bin15_Triboson
490 prop_binCROS_electron2018_bin15_ZZtoLep
491 prop_binCROS_electron2016M_bin6_TTTo2L2Nu
492 prop_binCROS_electron2016M_bin17_TVX
493 prop_binCROS_electron2017_bin11_Triboson
494 prop_binCROS_muon2017_bin8_Triboson
495 prop_binCROS_muon2017_bin8_TVX
496 prop_binCROS_muon2017_bin9_ZZtoLep
497 prop_binCROS_muon2017_bin8_ZZtoLep
498 lep_2016M
499 prop_binCROS_muon2017_bin4_WZ
500 prop_binCROS_muon2017_bin4_VG
501 prop_binCROS_muon2017_bin4_Triboson
502 prop_binCROS_muon2017_bin4_ZZtoLep
503 prop_binCROS_electron2018_bin15_Fake
504 prop_binCROS_electron2017_bin11_TVX
505 prop_binCROS_muon2018_bin6_ZZtoLep
506 prop_binCROS_muon2018_bin6_Triboson
507 prop_binCROS_electron2016M_bin6_ZZtoLep
508 prop_binCROS_electron2016M_bin6_TVX
509 prop_binCROS_electron2017_bin9_Triboson
510 prop_binCROS_electron2017_bin9_ZZtoLep
−2 −1 0 1 2
−0.01 0 0.01
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.32}_{-0.10}$
511 prop_binCROS_electron2018_bin15_TTTo2L2Nu
512 prop_binCROS_muon2017_bin13_TVX
513 prop_binCROS_electron2016M_bin9_TVX
514 prop_binCROS_electron2016M_bin9_ZZtoLep
515 prop_binCROS_electron2017_bin11_ZZtoLep
516 prop_binSR_muon2017_bin9_Triboson
517 prop_binCROS_muon2016M_bin14
518 prop_binCROS_muon2018_bin6_TVX
519 prop_binSR_electron2017_bin13_WrongSign
520 FES
521 prop_binSR_electron2016M_bin2_WpWpJJ_QCD
522 prop_binCROS_electron2016M_bin15_Fake
523 prop_binSR_electron2016M_bin2_Triboson
524 QCDScale_sig
525 prop_binCROS_electron2016M_bin15_TTTo2L2Nu
526 VBS_2018
527 prop_binSR_electron2018_bin16_WpWpJJ_EWK
528 prop_binCROS_electron2018_bin15_WpWpJJ_EWK
529 prop_binCROS_electron2018_bin14_Triboson
530 prop_binCROS_muon2017_bin4_TVX
531 prop_binCROS_electron2016M_bin7_Triboson
532 prop_binCROS_electron2016M_bin6_Triboson
533 prop_binCROS_electron2016M_bin6_VG
534 prop_binCROS_electron2018_bin10_TVX
535 VBS_2017
536 prop_binCROS_electron2016M_bin9_Triboson
537 prop_binCROS_muon2016M_bin9_Triboson
538 prop_binCROS_electron2017_bin9_TVX
539 prop_binSR_electron2017_bin17_WpWpJJ_EWK
540 prop_binCROS_muon2017_bin12_WpWpJJ_EWK
−2 −1 0 1 2
−10 −5 0 5 10
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = -0.00 +0.32 −0.10
541 QCDScale_WZ
542 VBS_2016M
543 prop_binCROS_electron2016M_bin15_WpWpJJ_EWK
544 prop_binCROS_electron2016M_bin17_WpWpJJ_EWK
545 prop_binCROS_electron2016M_bin6_WpWpJJ_EWK
546 prop_binCROS_electron2016M_bin7_WpWpJJ_EWK
547 prop_binCROS_electron2016M_bin9_WpWpJJ_EWK
548 prop_binCROS_electron2017_bin10_WpWpJJ_EWK
549 prop_binCROS_electron2017_bin9_WpWpJJ_EWK
550 prop_binCROS_electron2018_bin10_WpWpJJ_EWK
551 prop_binCROS_electron2018_bin14_WpWpJJ_EWK
552 prop_binCROS_muon2016M_bin9_WpWpJJ_EWK
553 prop_binCROS_muon2017_bin13_WpWpJJ_EWK
554 prop_binCROS_muon2017_bin4_WpWpJJ_EWK
555 prop_binCROS_muon2017_bin8_WpWpJJ_EWK
556 prop_binCROS_muon2017_bin9_WpWpJJ_EWK
557 prop_binCROS_muon2018_bin6_WpWpJJ_EWK
558 prop_binSR_electron2016M_bin16_WpWpJJ_EWK
559 prop_binSR_electron2016M_bin17_WpWpJJ_EWK
560 prop_binSR_electron2016M_bin2_WpWpJJ_EWK
561 prop_binSR_electron2017_bin13_WpWpJJ_EWK
562 prop_binSR_muon2016M_bin16_WpWpJJ_EWK
563 prop_binSR_muon2017_bin14_WpWpJJ_EWK
564 prop_binSR_muon2017_bin16_WpWpJJ_EWK
565 prop_binSR_muon2017_bin9_WpWpJJ_EWK
566 tau_vsele_2016M
567 tau_vsele_2017
568 tau_vsele_2018
569 tau_vsmu_2016M
570 tau_vsmu_2017
−2 −1 0 1 2
−1 −0.5 0 0.5 1
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = -0.00$^{+0.32}_{-0.10}$
571
tau_vsmu_2018
−2 −1 0 1 2
−1 −0.5 0 0.5 1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$